RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Thursday, 27 August 2026, 10:46 AM
State	Finished
Completed on	Thursday, 27 August 2026, 10:47 AM
Time taken	1 min 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int countOccurrences(int nums[], int left, int right, int target)
{
    int count = 0;

    for (int i = left; i <= right; i++)
    {
        if (nums[i] == target)
            count++;
    }

    return count;
}

int majorityElement(int nums[], int left, int right)
{

    if (left == right)
        return nums[left];

    int mid = left + (right - left) / 2;

    int leftMajority = majorityElement(nums, left, mid);
    int rightMajority = majorityElement(nums, mid + 1, right);

    if (leftMajority == rightMajority)
        return leftMajority;

    int leftCount = countOccurrences(nums, left, right, leftMajority);
    int rightCount = countOccurrences(nums, left, right, rightMajority);

    if (leftCount > rightCount)
        return leftMajority;
    else
        return rightMajority;
}

int main()
{
    int n;
```

```
42
43      scanf("%d", &n);
44
45      int nums[n];
46
47      for (int i = 0; i < n; i++)
48          scanf("%d", &nums[i]);
49
50      int result = majorityElement(nums, 0, n - 1);
51
52      printf("%d\n", result);
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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